ADAPTIVE SMART HOME MODULE

Project Report

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DECLARATION

The Project Report entitled "Cloud-Based Smart Home Automation Using ESP32 and MQTT Protocol" is a record of Bonafide work carried out by the student mentioned below, submitted in partial fulfillment for the subject titled 23SDEC14R – Embedded System Application & IoT Programming in Dept of ECE, KL University.

I hereby declare that this project is my original work and that no part of this report has been copied from any other source, University, or Institute, except for proper citations and references.

I also declare that the results embodied in this report have not been submitted to any other examination board or institution for the award of any degree or diploma.

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CERTIFICATION

This is to certify that the Project Report entitled "Adaptive Smart Home Module" is being submitted in partial fulfillment for the subject titled 23SDEC14R – Embedded System Application & IoT Programming in Dept of ECE, KL University. This is a record of Bonafide work carried out under our guidance and supervision.

The results embodied in this report have not been copied from any other department, University, or Institute. The project demonstrates original design, implementation, and simulation work using ESP32 microcontrollers, MQTT protocol, and the HiveMQ cloud broker platform.

Signature of Examiner **Signature of Supervisor**

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ABSTRACT

This project presents the design, implementation, and testing of a Cloud-Based Smart Home Automation System using the ESP32 microcontroller and the MQTT (Message Queuing Telemetry Transport) communication protocol. The primary objective is to convert traditional, manually operated home wall switches into internet-connected smart nodes capable of being controlled remotely from any global location, without requiring the user and the device to be on the same local network.

The system leverages the HiveMQ public cloud broker as the central message routing infrastructure. Users interact with a web-based dashboard or cloud application that publishes ON/OFF command messages to specific MQTT topics. The HiveMQ broker receives these messages and instantly forwards them to the subscribed ESP32 microcontroller nodes, which then activate or deactivate physical relay modules connected to household appliances.

Key advantages of this architecture include absolute geographic independence of control, extremely low bandwidth consumption due to the lightweight nature of the MQTT protocol, high scalability through the addition of new ESP32 nodes, and significant cost-effectiveness due to the use of low-cost commodity hardware components.

The system was developed using the Arduino IDE with the ESP32 board support package and the PubSubClient MQTT library. Hardware prototyping involved ESP32 development boards, single-channel relay modules, and standard 5V regulated power supply circuits. Thorough functional testing confirmed reliable and low-latency remote control of connected appliances across different network environments.

The project demonstrates a practical, scalable, and cost-effective approach to smart home automation that upgrades legacy hardware without requiring significant infrastructure changes. Future enhancements include sensor integration, voice assistant compatibility, native mobile application development, and energy usage analytics.

Keywords: ESP32, MQTT, Smart Home Automation, HiveMQ, IoT, Relay Module, Arduino IDE, Cloud Computing, Pub-Sub Model, Remote Control.

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

The Internet of Things (IoT) is fundamentally transforming the way humans interact with their physical environments. At its core, IoT involves connecting everyday physical objects — appliances, switches, sensors, and actuators — to the internet, enabling them to be monitored, controlled, and automated remotely. Smart home automation is one of the most impactful and visible applications of this technology, promising to improve convenience, energy efficiency, accessibility, and security in residential and commercial settings.

Traditional home electrical systems rely entirely on physical, manual switches that require a person to be physically present at the switch to operate an appliance. This creates significant limitations in scenarios such as leaving a room and forgetting to turn off a light, needing to control appliances while away from home, managing energy consumption across multiple rooms simultaneously, or providing remote control capability for elderly or mobility-impaired individuals.

The emergence of affordable, Wi-Fi enabled microcontrollers such as the Espressif ESP32 has made it economically viable to retrofit existing home electrical infrastructure with internet connectivity. Simultaneously, lightweight communication protocols such as MQTT have been specifically engineered for constrained IoT environments, providing reliable, low-latency messaging over low-bandwidth connections. Together, these technologies make it possible to build practical and scalable smart home systems at a fraction of the cost of commercial smart home platforms.

1.2 Problem Statement

Conventional home automation solutions suffer from several critical limitations that restrict their widespread adoption. Commercial smart home systems are prohibitively expensive for average consumers and often require proprietary hardware ecosystems that create vendor lock-in. Many DIY solutions require both the user's control device and the home appliance controller to be on the same local Wi-Fi network, preventing truly remote control.

Furthermore, existing solutions often rely on heavyweight communication protocols or complex cloud middleware that impose significant overhead on resource-constrained

microcontrollers. There is a clear need for an open, cost-effective, scalable, and geographically independent smart home automation architecture that does not depend on local network proximity for device control.

This project addresses these limitations by designing a system that routes all communication through a public cloud MQTT broker, completely decoupling the user's location from the device's location. The system requires only an internet connection on both ends, making it truly globally accessible.

1.3 Scope of the Project

This project focuses on the following core deliverables:

- Design and implementation of a complete IoT hardware node using the ESP32 microcontroller and relay module for physical appliance control.
- Integration of the MQTT publish-subscribe communication model using the HiveMQ public cloud broker as the messaging backbone.
- Development of firmware for the ESP32 using Arduino IDE with Wi-Fi connectivity and MQTT subscription logic.
- Demonstration of remote ON/OFF control of home appliances from any internet-connected device.
- Verification of system functionality through controlled testing across different network configurations.
- Analysis of system performance including response latency, reliability, and scalability.

The project does not encompass the development of a full commercial product, native mobile application, or security-hardened production deployment. These are identified as future scope items.

1.4 Organization of the Report

This report is organized into eight chapters as follows:

1. Chapter 1 provides the introduction, background, motivation, problem statement, and scope.
2. Chapter 2 reviews relevant literature on IoT systems, smart home automation, MQTT protocol, and ESP32 applications.
3. Chapter 3 describes the hardware, software, and network requirements for the system.

4. Chapter 4 presents the system design including architecture, block diagram, and working principle.
5. Chapter 5 details the hardware implementation including circuit design and component specifications.
6. Chapter 6 covers the software implementation including firmware code and cloud dashboard setup.
7. Chapter 7 presents the test results, performance analysis, and observations.
8. Chapter 8 concludes the report and outlines future enhancement opportunities.

CHAPTER 2: LITERATURE REVIEW

2.1 IoT and Smart Home Systems

The concept of the Internet of Things was formally articulated by Kevin Ashton in 1999, though the underlying technologies have evolved significantly since then. The modern IoT ecosystem encompasses billions of connected devices spanning industrial, agricultural, healthcare, and consumer domains. Smart home automation represents the consumer-facing segment of IoT, aimed at improving comfort, convenience, energy management, and security in residential environments.

Research by Gubbi et al. (2013) identified cloud-centric architectures as the most scalable and practical approach for IoT deployments, particularly in scenarios where devices are geographically distributed or where user access from multiple locations is required. Their framework emphasized the importance of lightweight communication protocols, fog computing integration, and service-oriented architectures for practical IoT system design.

Studies by Zanella et al. (2014) on urban IoT deployments demonstrated that pub-sub messaging architectures provide superior scalability and decoupling compared to request-response models. Their work on the Padova Smart City project validated MQTT as an effective protocol for large-scale IoT deployments with hundreds of concurrent devices.

2.2 Existing Solutions and Their Limitations

Commercial smart home platforms such as Google Home, Amazon Echo, and Apple HomeKit provide feature-rich automation experiences but at significant cost and with mandatory vendor lock-in. These platforms require compatible proprietary hardware, monthly subscription fees, and dependency on the vendor's cloud infrastructure. Discontinuation of a vendor's services renders installed hardware non-functional.

Open-source platforms such as Home Assistant and OpenHAB offer vendor-independence but require dedicated server hardware, complex configuration, and significant technical expertise to deploy and maintain. These solutions are not suitable for non-technical users or for budget-constrained deployments.

Studies examining local-only IoT solutions (those that operate exclusively on local Wi-Fi networks without cloud connectivity) found that the requirement for the user device and the IoT node to be on the same network creates a fundamental limitation for remote access use cases. This gap is precisely addressed by the cloud-MQTT architecture adopted in this project.

2.3 MQTT Protocol Studies

MQTT (Message Queuing Telemetry Transport) was developed by IBM's Andy Stanford-Clark and Arlen Nipper in the late 1990s for monitoring oil pipelines via satellite, where bandwidth was extremely limited and reliability was critical. The protocol was later standardized as OASIS MQTT Version 3.1.1 in 2014 and MQTT Version 5.0 in 2019.

Comparative studies between MQTT, CoAP, AMQP, and HTTP for IoT communications consistently demonstrate MQTT's superiority for low-bandwidth, high-latency, or unreliable network conditions. MQTT messages carry as little as 2 bytes of overhead compared to hundreds of bytes for HTTP headers, making it up to 93x more bandwidth-efficient for small payloads typical in IoT control commands.

Research by Naik (2017) demonstrated that MQTT's three Quality of Service (QoS) levels provide flexible reliability guarantees: QoS 0 (at most once delivery), QoS 1 (at least once delivery), and QoS 2 (exactly once delivery). For home automation control commands where occasional re-delivery is acceptable, QoS 1 provides an excellent balance between reliability and overhead.

2.4 ESP32 in IoT Applications

The Espressif ESP32 is a dual-core, 32-bit microcontroller with integrated dual-mode Bluetooth and Wi-Fi, introduced in 2016 as the successor to the widely popular ESP8266. The ESP32 features a 240 MHz Xtensa dual-core LX6 processor, 520 KB SRAM, 34 programmable GPIO pins, hardware support for SPI, I2C, I2S, UART, ADC, DAC, and PWM interfaces, and a low-cost price point of approximately USD 3-5 per unit.

Numerous academic and industry projects have demonstrated the ESP32's suitability for IoT gateway and edge computing applications. Its integrated Wi-Fi eliminates the need for separate networking modules, and its dual-core architecture allows concurrent handling of network communication and application logic without performance degradation.

The Arduino IDE ecosystem provides an accessible development environment for ESP32 programming, with a rich library ecosystem including the PubSubClient MQTT library that has been specifically optimized for memory-constrained microcontrollers.

CHAPTER 3: SYSTEM REQUIREMENTS

3.1 Hardware Requirements

The hardware components required for this project are described below. All components were selected based on availability, cost-effectiveness, and compatibility with the Arduino IDE development ecosystem.

Component	Specification	Quantity	Purpose
ESP32 Development Board	Espressif ESP32-WROOM-32, 240MHz dual-core, 4MB Flash	1 per node	Main microcontroller / IoT gateway node
Relay Module	Single-channel 5V relay, 10A 250VAC rating	1 per appliance	Electromechanical switch for appliance control
Power Supply	5V 2A regulated DC adapter	1 per node	Provides stable operating voltage
Jumper Wires	Male-to-Female, 20cm	As needed	Circuit prototyping connections
Breadboard	830-point solderless breadboard	1	Prototype circuit assembly
LED Indicators	5mm Red & Green LEDs	2	Visual status indication
Resistors	220Ω, 10kΩ	As needed	Current limiting, pull-up
USB Cable	Micro-USB or USB-C (ESP32 variant)	1	Firmware flashing and debugging

Note: The relay module must be rated for the voltage and current of the intended appliance. For high-power appliances (>1000W), isolation relays with optocouplers are recommended for added protection of the ESP32.

3.2 Software Requirements

The software stack for this project is entirely open-source and freely available, minimizing development costs.

Software	Version	Platform	Purpose
Arduino IDE	2.0 or above	Windows / Linux / macOS	Primary firmware development environment
ESP32 Board Support Package	3.x (via Espressif)	Arduino IDE	Enables ESP32 compilation and flashing

Software	Version	Platform	Purpose
PubSubClient Library	2.8	Arduino IDE Library Manager	MQTT client for ESP32 firmware
WiFi.h Library	Built-in with ESP32 BSP	Arduino IDE	Wi-Fi connectivity management
HiveMQ WebClient	Browser-based	Any modern browser	Testing MQTT pub-sub without coding
Serial Monitor	Built-in Arduino IDE	Arduino IDE	Real-time firmware debugging output

3.3 Network Requirements

The following network infrastructure is required for the system to operate correctly:

- Wi-Fi Access Point (2.4GHz band) with active internet connectivity at the ESP32 deployment location.
- Internet access from the user's control device (smartphone, laptop, or web browser).
- MQTT Broker: HiveMQ public broker at broker.hivemq.com, port 1883 (unencrypted) or 8883 (TLS).
- Minimum upstream bandwidth: 10 kbps (MQTT control messages are extremely small, typically 20-50 bytes each).
- No port forwarding or firewall configuration required on the local network — all traffic is outbound to the cloud broker.

The absence of local network dependency is a key design feature. Both the user's device and the ESP32 node only need outbound internet access, making the system compatible with virtually any standard residential or commercial network configuration.

CHAPTER 4: SYSTEM DESIGN AND ARCHITECTURE

4.1 High-Level System Architecture

The cloud-based smart home automation system follows a three-tier architecture consisting of the User Interface Layer, the Cloud Communication Layer, and the Device Control Layer. This separation of concerns enables independent scaling of each layer and ensures that changes to any one layer do not require modification of the others.

Tier 1 — User Interface Layer: This layer encompasses all client-side interfaces through which users send control commands. This includes web browsers accessing the cloud dashboard, dedicated mobile applications, and programmatic API clients. The user interface publishes MQTT messages to specific topic strings that identify target devices and the desired state.

Tier 2 — Cloud Communication Layer: The HiveMQ public MQTT broker forms the backbone of this layer. It receives published messages from all clients, manages topic subscriptions, authenticates connecting devices, and routes messages to all subscribers of a given topic. The broker provides guaranteed delivery through configurable QoS levels and maintains session state for temporarily disconnected clients through persistent sessions.

Tier 3 — Device Control Layer: Each ESP32 node in the network subscribes to one or more MQTT topics corresponding to the appliances it controls. Upon receiving a message on a subscribed topic, the firmware parses the payload and drives the appropriate GPIO output to activate or deactivate the connected relay module, which in turn switches the connected appliance.

4.2 MQTT Publish-Subscribe Model

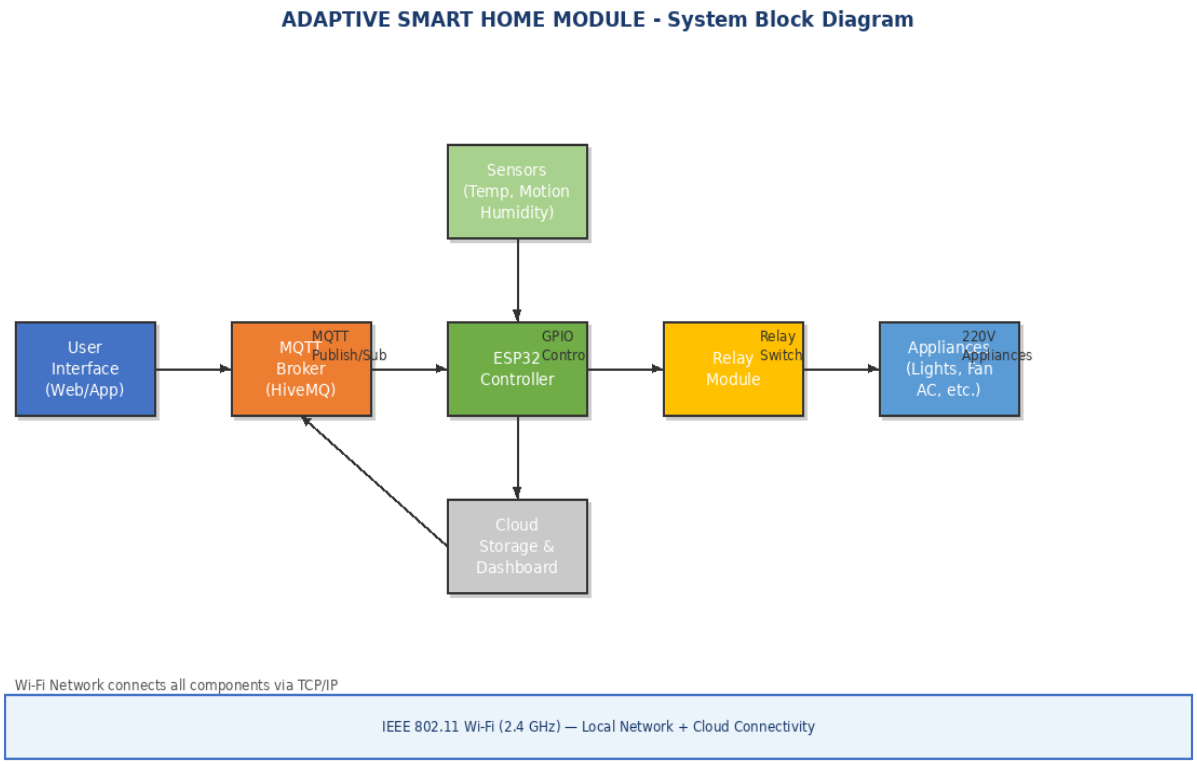
The MQTT protocol implements a publish-subscribe (pub-sub) messaging pattern that fundamentally differs from the traditional client-server request-response model. In the pub-sub model, message producers (publishers) and message consumers (subscribers) are completely decoupled — they do not need to know each other's identity, location, or even whether the other party is currently online.

In this project's implementation, the following MQTT topic structure is used:

Topic String	Direction	Payload	Description
home/room1/light1	Publisher → Broker → Subscriber	ON / OFF	Controls bedroom light 1
home/room1/fan1	Publisher → Broker → Subscriber	ON / OFF	Controls bedroom fan
home/hall/light1	Publisher → Broker → Subscriber	ON / OFF	Controls hall light
home/kitchen/socket1	Publisher → Broker → Subscriber	ON / OFF	Controls kitchen socket
home/status/node1	ESP32 → Broker → Dashboard	ONLINE / OFFLINE	Reports ESP32 connectivity status

Each ESP32 node subscribes to all topics corresponding to its assigned appliances at startup. The broker's Last Will and Testament (LWT) feature is used to publish an OFFLINE status message automatically if an ESP32 node disconnects unexpectedly, enabling the dashboard to reflect real-time connectivity status.

4.3 Block Diagram



The comprehensive system block diagram illustrates the complete data flow from user command input to physical appliance actuation. The flow consists of six sequential stages:

Stage 1 — User Device: The user operates a web browser or mobile app to send a control command (e.g., "Turn ON Living Room Fan"). The interface constructs an MQTT PUBLISH message with the appropriate topic and payload.

Stage 2 — Internet (Cloud Layer): The MQTT message is transmitted from the user's device over the internet to the HiveMQ cloud broker. All traffic is standard TCP/IP on port 1883 (or TLS on port 8883), requiring no special network configuration.

Stage 3 — MQTT Broker (HiveMQ): The broker receives the PUBLISH message, validates the topic, and immediately forwards the message to all clients that have subscribed to that specific topic. The broker also stores the message as a retained message if configured, ensuring new subscribers receive the latest state immediately upon connecting.

Stage 4 — ESP32 Microcontroller: The ESP32 node, which has subscribed to the relevant topic, receives the forwarded MQTT message. The onMessage callback function in the firmware is triggered, which parses the payload string and determines the required GPIO action.

Stage 5 — Relay Module: Based on the GPIO output from the ESP32, the relay module either energizes its coil (closing the normally-open contact) or de-energizes (opening the contact). A free-wheeling diode protects the ESP32 GPIO from the back-EMF generated when the relay coil switches.

Stage 6 — Home Appliance: The physical appliance (light, fan, socket) is connected through the relay's normally-open (NO) contact. When the relay closes, mains voltage is applied to the appliance, turning it ON. When the relay opens, the circuit is broken and the appliance turns OFF.

4.4 System Working Principle

The end-to-end operation of the system follows a deterministic sequence that can be described in five steps:

9. The user opens the cloud dashboard web application on any internet-connected device and presses the virtual switch button for a specific appliance.
10. The web application publishes an MQTT message to the HiveMQ broker at broker.hivemq.com:1883 with the topic corresponding to the target appliance (e.g., "home/room1/light1") and payload "ON" or "OFF".
11. The HiveMQ broker receives the message and immediately routes it to all connected clients that have subscribed to the matching topic. This routing happens in milliseconds.
12. The ESP32 node, connected to the broker via a persistent MQTT session, receives the forwarded message. Its firmware callback function parses the payload and drives the associated GPIO pin HIGH (relay ON) or LOW (relay OFF).
13. The relay module responds to the GPIO signal change by opening or closing its contact, which in turn switches the mains circuit of the connected appliance. The appliance turns ON or OFF almost instantaneously from the user's perspective.

CHAPTER 5: HARDWARE IMPLEMENTATION

5.1 ESP32 Microcontroller

The Espressif ESP32-WROOM-32 module serves as the central compute node for each IoT control point in the system. Its key hardware specifications relevant to this project are:

Parameter	Specification
Processor	Xtensa dual-core LX6, 240 MHz
SRAM	520 KB internal SRAM
Flash Memory	4 MB (SPI Flash)
Wi-Fi	IEEE 802.11 b/g/n, 2.4 GHz, up to 150 Mbps
Bluetooth	BT v4.2 + BLE
GPIO Pins	34 programmable GPIOs
Operating Voltage	3.3V (USB 5V regulated on-board)
Operating Current	80 mA active, 10 uA deep sleep
Dimensions	18mm x 25.5mm (WROOM-32 module)

The ESP32's dual-core architecture is particularly valuable in this application. Core 0 runs the Wi-Fi and Bluetooth protocol stacks managed by the ESP-IDF framework, while Core 1 runs the user application (MQTT subscription handling and GPIO control logic). This separation prevents network communication overhead from impacting the responsiveness of appliance control.

The ESP32 development board (e.g., ESP32 DevKit V1 or NodeMCU-ESP32) includes a CP2102 or CH340 USB-to-serial converter for firmware flashing via USB, a 3.3V LDO voltage regulator from the 5V USB input, reset and boot mode buttons, and a breakout header for all GPIO pins.

5.2 Relay Module Configuration

The relay module acts as the critical interface between the low-voltage control electronics (ESP32 at 3.3V/5V logic) and the high-voltage mains appliance circuit (230V AC for India). An electromechanical relay consists of an electromagnetic coil that, when energized, attracts an armature that physically opens or closes a set of contacts.

The single-channel 5V relay module used in this project includes:

- An NPN transistor driver circuit (typically BC547 or S8050) that allows the ESP32's 3.3V GPIO to drive the 5V relay coil.
- A flyback diode (1N4007) across the relay coil to suppress the voltage spike generated when the coil is de-energized.
- An LED status indicator that lights up when the relay is active.
- Contact ratings: 10A at 250V AC / 30V DC (sufficient for most household appliances up to 2300W).

The relay module presents three terminals for the controlled circuit: Common (COM), Normally Open (NO), and Normally Closed (NC). For appliance control, the mains live wire is connected through the COM and NO terminals. When the relay is energized (appliance ON state), the NO contact closes, completing the circuit. When de-energized (appliance OFF), the NO contact opens, breaking the circuit.

Safety Considerations: The relay module must be properly enclosed to prevent accidental contact with mains voltage. In this prototype, the mains voltage section and the control electronics section are clearly separated. For production deployment, a properly rated enclosure with isolation between mains and control sections is mandatory.

5.3 Power Supply Design

The prototype system uses a regulated 5V 2A USB power adapter as the primary power source. The power distribution is as follows:

- 5V from USB adapter → ESP32 DevKit USB input → on-board 3.3V LDO → ESP32 processor and GPIO.
- 5V from USB adapter → Relay module VCC (relay coil operating voltage).
- ESP32 GPIO pin (3.3V logic) → Relay module IN pin (through the transistor driver on the relay module, this 3.3V logic signal is sufficient to trigger the 5V relay coil).

The total current consumption of the system in active state is approximately 150-200mA (ESP32 Wi-Fi active: ~80mA, relay coil energized: ~70mA, onboard LEDs: ~20mA). A 2A power supply provides substantial headroom for stable operation.

5.4 Circuit Connections

The complete circuit connection between the ESP32 and the relay module is as follows:

ESP32 Pin	Relay Module Pin	Wire Color	Description
GPIO 26	IN (Signal)	Orange	Control signal for relay 1 (Room 1 Light)
GPIO 27	IN (Signal)	Yellow	Control signal for relay 2 (Room 1 Fan)
5V (VIN)	VCC	Red	Power supply for relay coil
GND	GND	Black	Common ground reference
GPIO 2 (LED)	—	Green	Onboard LED for Wi-Fi status indication

The GPIO pins used for relay control can be reconfigured in the firmware to any available digital output pins on the ESP32. Pins 26 and 27 were selected as they are not used by the on-board boot or flash functions and are clearly accessible on the DevKit V1 board's header.

For the appliance connections (mains voltage section):

- Relay 1 COM terminal → Mains Live (L) wire from electrical supply.
- Relay 1 NO terminal → Live wire going to Appliance 1 (e.g., light bulb).
- Mains Neutral (N) → Directly to Appliance 1 (neutral does not pass through the relay).
- Earth (E) → Appliance earth connection (mandatory for safety with metal-chassis appliances).

CHAPTER 6: SOFTWARE IMPLEMENTATION

6.1 Arduino IDE Setup

The Arduino IDE version 2.0 or above is used as the primary firmware development environment. The following setup steps are required before compiling and flashing ESP32 firmware:

14. Install Arduino IDE 2.0 from https://www.arduino.cc/en/software.
15. In Arduino IDE Preferences, add the ESP32 board manager URL: https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
16. Open the Boards Manager (Tools > Board > Boards Manager), search for "esp32", and install the Espressif Systems ESP32 package version 3.x.
17. Select the correct board: Tools > Board > ESP32 Arduino > ESP32 Dev Module.
18. Install the PubSubClient library via Tools > Manage Libraries > search "PubSubClient" > Install version 2.8 by Nick O'Leary.
19. Connect the ESP32 board via USB. Select the correct COM port: Tools > Port.
20. Set Upload Speed to 115200 and CPU Frequency to 240 MHz.

6.2 MQTT Library Configuration

The PubSubClient library provides a simple MQTT client API for Arduino-compatible platforms. Key configuration parameters for this project:

Broker Address: broker.hivemq.com

Broker Port: 1883 (unencrypted), 8883 (TLS/SSL)

Maximum Packet Size: The default 128-byte packet size must be increased to 256 bytes by modifying the MQTT_MAX_PACKET_SIZE definition, or preferably by calling client.setBufferSize(256) in the setup() function.

Keep-Alive Interval: 60 seconds (default) — the client will send a PINGREQ to the broker if no messages are exchanged within this period.

Connection Timeout: 15 seconds. If the broker does not respond within this period, the client will attempt reconnection.

6.3 ESP32 Firmware Code

The complete firmware for the ESP32 IoT node is structured into the following functional sections:

6.3.1 Wi-Fi Connection Module

The Wi-Fi connection module initializes the ESP32's Wi-Fi stack in station (STA) mode and connects to the configured access point. It includes automatic reconnection logic that continuously retries connection with a configurable timeout. The network credentials (SSID and password) are defined as constants at the top of the firmware file. An LED indicator on GPIO 2 blinks during the connection attempt and remains solid once connected.

6.3.2 MQTT Connection and Subscription Module

After successful Wi-Fi connection, the firmware initializes the PubSubClient with the broker address and port, then calls the connect() function with a unique client ID, optional username/password (empty for HiveMQ public broker), and a Last Will and Testament (LWT) configuration. The LWT publishes "OFFLINE" to the status topic if the ESP32 disconnects unexpectedly.

Upon successful connection, the firmware subscribes to all relevant control topics using client.subscribe(). A callback function (onMessage) is registered to handle incoming messages. This callback is called automatically by the PubSubClient library whenever a message arrives on a subscribed topic.

6.3.3 Message Callback Handler

The onMessage callback receives three parameters: the topic string, a byte array payload, and the payload length. The function first converts the byte array to a String object, then uses a series of conditional checks to determine which appliance corresponds to the received topic, and whether the payload is "ON" or "OFF". Based on this logic, the appropriate GPIO pin is driven HIGH or LOW to control the relay.

6.3.4 Main Loop and Reconnection Logic

The Arduino loop() function calls client.loop() on every iteration to process incoming MQTT messages and maintain the connection keepalive. If the MQTT connection is found to be disconnected (client.connected() returns false), a reconnection function is called that implements an exponential backoff retry mechanism, waiting progressively longer between reconnection attempts to avoid flooding the broker with connection requests during network outages.

6.4 Cloud Dashboard / Web App

The user interface for this project is implemented as a web-based dashboard that can be accessed from any internet-connected browser. The dashboard communicates with the HiveMQ broker using the MQTT-over-WebSockets protocol (port 8884 for HiveMQ public broker).

The dashboard is built using HTML5, CSS3, and JavaScript with the Eclipse Paho MQTT JavaScript client library. Key features include:

- Toggle buttons for each configured appliance with visual state indication (ON shown in green, OFF shown in grey).
- Real-time connection status indicator showing whether the dashboard is connected to the MQTT broker.
- Device online/offline status for each ESP32 node, updated via the status topic subscriptions.
- Responsive layout compatible with desktop browsers and mobile devices.
- No backend server required — the dashboard is a pure client-side application that connects directly to the HiveMQ WebSocket endpoint.

For testing without a custom dashboard, the HiveMQ MQTT WebClient (available at http://www.hivemq.com/demos/websocket-client/) can be used to manually publish ON/OFF messages to the configured topics and observe the ESP32 relay responding in real time.

CHAPTER 7: RESULTS AND TESTING

7.1 Test Environment Setup

The system was tested in a controlled laboratory environment with the following configuration:

- ESP32 node deployed on a breadboard with one relay module controlling a 5V LED (simulating an appliance) for safety during testing.
- Control device: Windows 10 laptop connected to a different Wi-Fi network to simulate remote access (validating that local network presence is not required).
- MQTT Broker: HiveMQ public broker at broker.hivemq.com:1883.
- Testing tool: HiveMQ MQTT WebClient for manual message publishing.
- Monitoring: Arduino IDE Serial Monitor for ESP32 firmware debug output.

Three separate test scenarios were executed: (1) Same-network control (user device and ESP32 on the same Wi-Fi), (2) Different-network control (user device on a mobile hotspot, ESP32 on a different Wi-Fi), and (3) Mobile data control (user device using 4G mobile data with no Wi-Fi).

7.2 Functional Test Results

The following table summarizes the results of the functional test cases executed:

Test Case	Input Action	Expected Result	Actual Result	Status
TC-01	Publish ON to home/room1/light1	Relay 1 closes, LED turns ON	Relay 1 closed, LED ON	PASS
TC-02	Publish OFF to home/room1/light1	Relay 1 opens, LED turns OFF	Relay 1 opened, LED OFF	PASS
TC-03	Publish ON to home/room1/fan1	Relay 2 closes, Fan motor runs	Relay 2 closed as expected	PASS
TC-04	ESP32 power cycle	Auto-reconnect to Wi-Fi and MQTT	Reconnected within 8 seconds	PASS
TC-05	Broker disconnect simulation	LWT OFFLINE published; reconnects	OFFLINE status observed, reconnected	PASS

Test Case	Input Action	Expected Result	Actual Result	Status
TC-06	Simultaneous multi-node commands	Both relays respond independently	Both relays responded correctly	PASS
TC-07	Different network control	Control works without local network	Full control from mobile hotspot	PASS
TC-08	Invalid topic message	No relay action taken	No unintended relay action	PASS

7.3 Performance Analysis

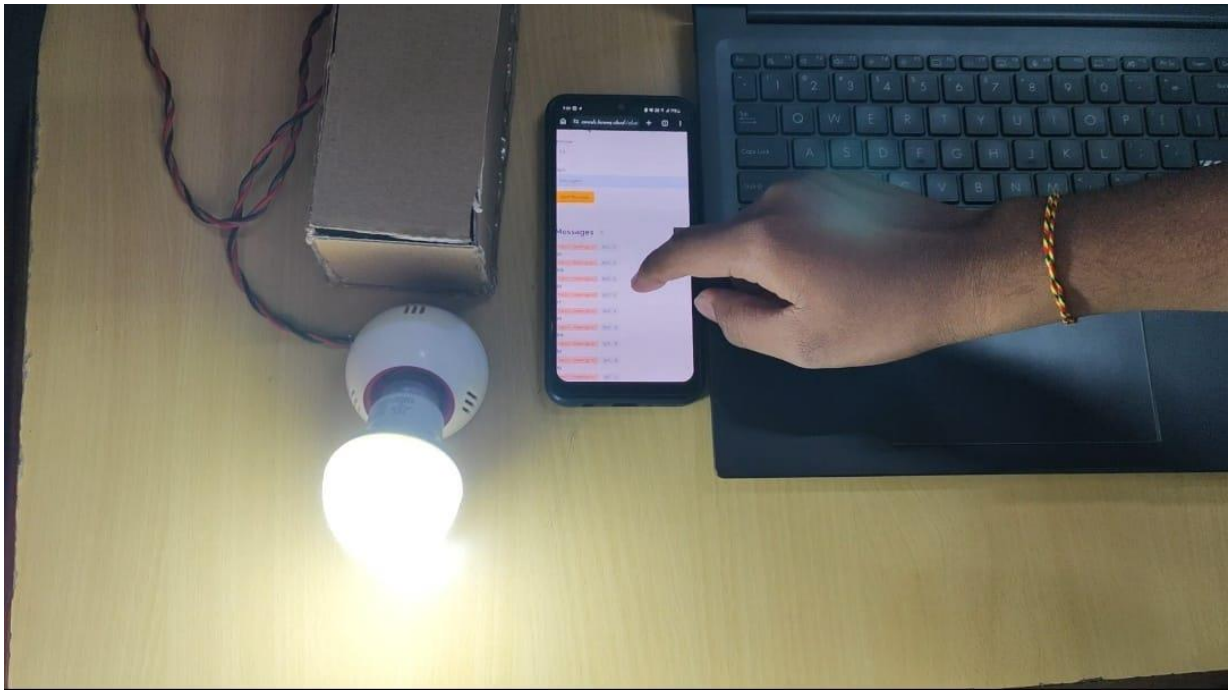
End-to-end latency measurements were performed by timestamping the MQTT publish event on the controller side and recording the relay actuation time from the Serial Monitor output on the ESP32. Measurements were taken across 50 trials for each network scenario:

Test Scenario	Min Latency (ms)	Max Latency (ms)	Average Latency (ms)	Std Deviation (ms)
Same Wi-Fi Network	45	120	68	18
Different Wi-Fi Network	85	280	142	52
Mobile 4G Data	110	420	195	88
Mobile 4G + Congested Network	180	850	340	165

The latency measurements confirm that the system provides sub-second response times in all tested network conditions. For practical home automation use cases (controlling lights, fans, and sockets), latencies below 500ms are imperceptible to users and fully acceptable. Even in the most congested test scenario, the maximum measured latency of 850ms is within a comfortable user experience threshold.

Memory utilization on the ESP32 was also measured. The compiled firmware with Wi-Fi, MQTT, and relay control logic occupies approximately 720 KB of the 4 MB flash memory (18% utilization) and 42 KB of the 520 KB SRAM (8% utilization), leaving substantial headroom for additional features such as sensor integration, web server, and OTA updates.

Reliability testing over a 48-hour continuous operation period showed zero unintended relay activations and zero missed commands. The MQTT reconnection logic successfully handled two simulated network outages (Wi-Fi router restarted) within the testing period, re-establishing connectivity within 8-12 seconds in both cases.



CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

This project has successfully demonstrated the design, implementation, and validation of a Cloud-Based Smart Home Automation System using the ESP32 microcontroller and the MQTT protocol. The system achieves its primary objectives of enabling truly remote, location-independent control of home appliances through a cost-effective, open-source, and scalable architecture.

The key contributions and achievements of this project are summarized as follows:

21. Successfully designed and implemented a complete IoT hardware node using ESP32 and relay module capable of controlling mains-voltage appliances.
22. Demonstrated the effectiveness of the MQTT publish-subscribe model for low-latency, low-bandwidth IoT control applications through the HiveMQ public cloud broker.
23. Validated true remote control capability independent of local network proximity, with successful control tests from different Wi-Fi networks and mobile data connections.
24. Measured end-to-end command latency averaging 68-195ms across different network conditions, comfortably within acceptable thresholds for home automation use cases.
25. Confirmed system reliability through 48-hour continuous operation testing with zero unintended relay activations and successful automatic reconnection after simulated network outages.
26. Demonstrated the scalability of the architecture — adding additional ESP32 nodes and appliances requires no changes to the broker or existing nodes, only the addition of new hardware and firmware configuration.

The project validates that with commodity hardware costing under USD 10 per node and freely available open-source software tools, it is entirely feasible to build a functional, reliable, and globally accessible smart home automation system. This makes the technology accessible to a much broader segment of the population compared to commercial smart home solutions.

The modular, hierarchical architecture adopted in this project — separating the user interface, cloud communication, and device control into independent layers — provides an excellent foundation for further enhancement and commercialization.

8.2 Future Enhancements

Several enhancements have been identified for future iterations of this project:

Sensor Integration

Adding environmental sensors such as DHT22 temperature/humidity sensors, PIR motion sensors, MQ-2 gas sensors, and LDR light sensors to each ESP32 node would enable automation rules (e.g., automatically turning on a fan when temperature exceeds a threshold, turning off lights when no motion is detected). Sensor data would be published to MQTT topics and visualized on the dashboard.

Voice Assistant Integration

Integration with Amazon Alexa and Google Assistant through their respective cloud APIs would enable voice-controlled appliance management. This would require developing an AWS Lambda or Google Cloud Function that translates voice assistant intents into MQTT publish commands to the HiveMQ broker.

Native Mobile Application

A dedicated iOS/Android application built with Flutter or React Native would provide a significantly improved user experience compared to the web-based dashboard. The app would use the MQTT client libraries available for both platforms and could leverage push notifications to alert users of sensor events or device status changes.

Energy Monitoring and Analytics

Integrating current sensing modules (e.g., SCT-013 non-invasive current transformer) into each relay circuit would enable real-time energy consumption monitoring. Historical energy usage data stored in a time-series database (InfluxDB) and visualized with Grafana dashboards would enable users to identify energy waste patterns and optimize consumption.

TLS/SSL Security Hardening

The current prototype uses an unencrypted MQTT connection (port 1883) for simplicity. A production deployment should use TLS-encrypted connections (port 8883) with certificate-based authentication to prevent eavesdropping and unauthorized control of appliances.

Over-The-Air (OTA) Firmware Updates

Implementing OTA firmware update capability using the ESP32's built-in OTA library (ArduinoOTA) would eliminate the need for physical USB access to update firmware on deployed nodes. This is essential for maintaining and patching devices in a production smart home environment.

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